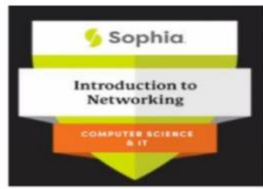


## CERTIFICATES



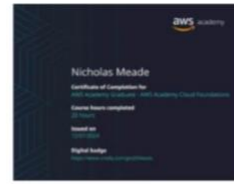
CompTIA A+



Sophia Networking



CompTIA Security+



AWS Cloud Fundamentals



Eureka Cybersecurity Graduate Certificate



Google Cybersecurity Certificate



SQL Training Certificate



Illinois Central College Linux Certificate



Python Core Certificate



Python AI Development Graduate Certificate



ISC2 CC (Certified In Cybersecurity)



CISCO Networking Badge



Illinois Central College Python Training Certificate

# PORTFOLIO PROJECTS

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## Project 1: IPv4 to Binary and Basic IPv4 Setup Explained

Everything in binary is in multiple of 2 different possible bits (0,1). There are 8 bits in a single IPv4 octet. So determining the value of the bit value place is always in the power of 2 (the 1 and 0, two different values) The first bit is always in the 1 place value  $2^0=1$ . The second place value is then  $2^1 (2 \times 1)$  which equals 2.  $2^2 (2 \times 2)=4$  and so forth for all 8 possible bits.  $2^3 (2 \times 2 \times 2)=8$ .  $2^4 (2 \times 2 \times 2 \times 2)=16$ .  $2^5=32$ .  $2^6=64$ .  $2^7=128$

Place value: 7 6 5 4 3 2 1 0  
Math Equation:  $2^7$   $2^6$   $2^5$   $2^4$   $2^3$   $2^2$   $2^1$   $2^0$   
Bit: 128 64 32 16 8 4 2 1 = 255 possible for an IP octet  
based on the 1 bit that trigger it, thus the ones below:  
1 1 1 1 1 1 1 1 = 8 bits total possible in an octet

Subnet is 255.255.255.0

11111111.11111111.11111111.0 total 24 1's in the network so the CIDR is /24(bits): no further room available

00000000 is the host so it is 8 bits, tho its not 1's, the 0's still take room so its still 8 bits because there is 8 zeros. But this is only 1 set of 8 0's instead of like the network set that has 3 sets of one's which is why the network portion is 24 bits. Even though the host still takes room with 8 bits, the 0's still signify that adjustments can be made per different device that is put on the network.

Commonly, this is what a device IP setting will look like:

IP: 192.168.0.5

|               |     |    |    |    |   |   |   |   |                             |
|---------------|-----|----|----|----|---|---|---|---|-----------------------------|
| Binary: (192) | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | a 1 is placed at every bit  |
|               | 1   | 1  | 0  | 0  | 0 | 0 | 0 | 0 | (128, 64, 32, 16, 8, 4, 2,  |
| (168)         | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | 1) that can fit into the    |
|               | 1   | 0  | 1  | 0  | 1 | 0 | 0 | 0 | octet address. For          |
| (0)           | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | Example: the first octet is |
|               | 0   | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 192. A 1 is placed at 128   |
| (5)           | 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 | 64. 128 + 64 = 192. Any     |
|               | 0   | 0  | 0  | 0  | 0 | 1 | 0 | 1 | more then that would go     |

over 192. Thus, the binary for 192 is 1 1 000000

(128) (64)=192

Final Binary = 11000000.10101000.00000000.00000101

192. 168. 0. 5.

Subnet: 255.255.255.0

Binary: 11111111.11111111.11111111.00000000 = 24 1's (thus the CIDR is 24), no more room (network portion must remain the same)

8 bit zero's at the end = can be changed per different host device.

The max bits in IPv4 is 32 bits. The CIDR (network portion) is 24 bits. The host portion is 8 bits.  $24 + 8 = 32$ .

With subset network, the max amount of host (the 8 zero's last section of the subset max) is 255 host ( $2^8$  (the 8 I'm referring to is the last 8 bits in the subnet)) which equals 255. thus filling the last section 11111111. Creating the entire network to be 11111111.11111111.11111111.11111111 (255.255.255.255) This is network capacity that will no longer function to over clogging.

To create a network that would allow more than 255 devices, you'd have to give more room for hosts. You can move to the third set of bit for host and create a CIDR /16 network which would look like this: 11111111.11111111.00000000.00000000 (255.255.0.0). So this would now be  $2^{16}$  which equals 65,532. This would allow over 65,532 host or devices on the network!

If you had an enterprise and needed even more host or devices, you could use the /8 CIDR which would then be 255.0.0.0. In binary, this would be 11111111.00000000.00000000.00000000. Now remember, bits go by 8's, So if the first CIDR is 24, then we went to 16 to create more 0's to use, the next would be 0 less thus creating more zero's and 8 less then 16 is 8. So the CIDR for the larges network you could make is CIDR. The CIDR 8 represents the 8 1's that are still there in the first set of bits. Now we have 24 0's. So to figure out how many devices we can use in our 0's we use the equation  $2^{24}$ . Remember, every thing is in the power of 2 because we got 1's and 0's. So  $2^{24}$  is a staggering 16,777,214 available hosts or devices on the network.

Gateway: 192.168.0.1 (As you can see, the gate way typically has the same address in the first 3 octet as a host device. The first 3 octet is the network address. The last octet is the host address. All devices on a network share the same network address. A gateway router is typically the first host on a network, hence its host ID is xxx.xxx.x.1).

DNS: 192.168.0.1 (DNS is needed so the network can convert web address to IP and vice versa but you need an ip for the dns server, which is usually the same as the gateway address unless you know the IP of an external DNS server.)

Final product:

IP: 192.168.0.5/24

Subset: 255.255.255.0

Gateway: 192.168.0.1

DNS: 192.168.0.1

## Project 2: TLS Encryption Explained: How HTTPS Protects User Data

When you visit a secure website (like <https://example.com>), your browser uses something called TLS—short for Transport Layer Security—to protect your data. You'll know it's working when you see the little lock icon in your browser's address bar.

TLS is the modern version of SSL, and it's the reason your logins, passwords, and personal info don't get exposed when you browse or shop online. It works by creating a secure 'tunnel' between your device and the website. That tunnel is just a fancy way of saying the important parts of your web traffic (like form data and cookies) are scrambled using encryption so that no one in the middle—like hackers or snooping Wi-Fi networks—can read it.

Now, here's where it gets interesting. Think of an internet packet like a shipping box. The outside of the box (which includes MAC addresses, IP addresses, and TCP port numbers) is like the FedEx label—it tells the network where to deliver it. That part has to stay visible. But inside that box is a locked chest, and that's the encrypted TLS data. Only the intended recipient—the real website—has the key to open it. Anyone along the delivery route can see where it's going, but they can't open the chest and see what's inside.

It's important to know that while your private data is protected, things like your IP address, the website's IP, and some routing info are still visible. That's how the internet knows where to send the traffic. But the actual content—like the stuff you type into login pages—is totally locked down.

TLS is used every day in cybersecurity. Without it, safe online shopping, banking, and even using social media wouldn't be possible. And while it's complex under the hood (involving certificates, encryption keys, and handshakes), the main idea is simple: only you and the website should know what's being said.

Learning how TLS works helped me better understand why encryption is so crucial in protecting users. It's one of those behind-the-scenes tools that powers nearly everything we trust online.

### Project 3: IAM Policies and Role-Based Access Control (RBAC) with Active Directory

IAM stands for Identity and Access Management. It's how organizations make sure the right people have the right access to the right resources—nothing more, nothing less. One of the most common tools used for this is Microsoft Active Directory (AD).

In Active Directory, users are organized into groups. Rather than giving permissions to individual users one by one, we assign permissions to groups. This is where RBAC, or Role-Based Access Control, comes in. RBAC means you create roles (like 'HR', 'IT Support', or 'Managers') and tie permissions to those roles. Then you simply add users to the right group based on their job role.

For example, instead of giving 'John in Accounting' permission to the payroll folder, you'd create a group called 'Payroll Access', give that group permission to the folder, and then add John to the group. Now if someone else joins the accounting team, all you have to do is add them to the group—no need to redo permissions from scratch.

IAM policies are the rules that define what users or groups are allowed to do. In Active Directory, these policies can be implemented through Group Policy Objects (GPOs) or built into access control lists. These policies help enforce security, reduce human error, and make access changes fast and scalable.

Learning how to set up AD groups and apply RBAC has helped me understand how large organizations manage security efficiently. It's a key part of cybersecurity that often goes unnoticed—but without it, data would be either locked up too tightly or left too wide open.

## Project 4: How VLANs Work and Their Role in Network Security

A VLAN, or Virtual Local Area Network, allows you to split one physical network into multiple logical networks. It's like drawing invisible lines inside a switch so that devices in one department can't talk to devices in another—unless explicitly allowed.

For example, you might put the HR department on VLAN 10, IT on VLAN 20, and Sales on VLAN 30. Even though they're all using the same switch and cables, VLANs make it so their traffic is separated. This improves performance and adds a layer of security by keeping sensitive traffic (like HR or Finance) isolated from the rest.

Setting up VLANs usually happens on the switch using the CLI (Command Line Interface). Here's a basic example of how to create and assign a VLAN on a Cisco switch:

```
configure terminal
vlan 10
name HR
exit
interface FastEthernet0/1
switchport mode access
switchport access vlan 10
exit
```

This creates VLAN 10, names it 'HR', and assigns port FastEthernet0/1 to it. Devices plugged into that port are now part of the HR VLAN.

VLANs are important in cybersecurity because they support network segmentation—a core security principle. By segmenting traffic, you reduce the attack surface. For example, if a device in Sales is compromised, that attack can't easily spread to HR or IT since they're on different VLANs.

Learning how to set up VLANs helped me understand how physical hardware can be logically divided for both performance and security. It's a smart way to manage and protect networks, especially in organizations with multiple departments.

## Project 5: GRC, Security Audits, and Industry Compliance Standards

GRC stands for Governance, Risk, and Compliance. It's a core part of cybersecurity and business operations that ensures organizations follow laws, manage risks, and maintain internal policies. GRC helps make sure companies are doing the right things, the right way, and staying secure while doing it.

A key part of GRC is security audits. These are formal checks that evaluate how well an organization is protecting its data and systems. Audits can be internal or external, and they often compare company practices against well-known security standards.

Some of the most common standards include:

- HIPAA – Health Insurance Portability and Accountability Act: Protects patient health data in healthcare environments.
- PCI-DSS – Payment Card Industry Data Security Standard: Ensures businesses handle credit card info securely.
- GDPR – General Data Protection Regulation: Governs how companies handle the personal data of EU citizens.

Security audits look for gaps, misconfigurations, and violations of these standards. If something doesn't match, it's documented and corrected. The goal is not just to pass an audit but to build trust with customers and avoid breaches or legal trouble.

Learning about GRC has shown me that cybersecurity isn't just technical—it's also about rules, documentation, and accountability. Every firewall or access control has a policy behind it, and understanding those policies helps connect the dots between IT and business goals.

## Project 6: OSI and TCP/IP Models: How Data Travels Through a Network

The OSI model and the TCP/IP model are like roadmaps that explain how data moves across a network. They break down the complex journey into layers, where each layer has a specific job. Understanding these models helps IT professionals troubleshoot issues and understand how technologies interact.

The OSI model has 7 layers:

1. Physical – cables, ports, and electrical signals.
2. Data Link – MAC addresses and Ethernet frames.
3. Network – IP addressing and routing.
4. Transport – reliable delivery with TCP or UDP.
5. Session – opening and managing communication sessions.
6. Presentation – data formatting and encryption.
7. Application – what the user sees (like a browser or app).

The TCP/IP model is simpler and more practical with 4 layers:

1. Network Interface – covers OSI layers 1 and 2.
2. Internet – same as OSI's Network layer.
3. Transport – same as OSI's Transport layer.
4. Application – combines OSI layers 5–7.

Even though these models are theoretical, they help explain things like why a website loads slowly (maybe a Transport issue) or why a device isn't found on a network (maybe a Network or Data Link issue).

Learning how to map network traffic to these layers helped me better understand how the internet works and how to break down complex network problems step by step.

## Project 7: How Next-Gen Firewalls, XDRs, and SIEMs Work Together

Modern cybersecurity tools like Next-Gen Firewalls (NGFWs), XDRs (Extended Detection and Response), and SIEMs (Security Information and Event Management) are designed to work together to detect, block, and respond to threats across the entire network.

A Next-Gen Firewall doesn't just block traffic by port or protocol—it can analyze traffic up to the application layer. This means it can detect things like unsafe web activity, malware downloads, or unauthorized access attempts, even if they come through a standard port. NGFWs often include intrusion prevention systems (IPS), identity-based access, and deep packet inspection.

XDR, or Extended Detection and Response, collects and correlates data from multiple sources—like endpoints, servers, email, and network tools—and ties them into a unified view. This helps security teams spot coordinated attacks. For example, if a user clicks a phishing link, the XDR can link that event to a strange login, file download, or malware detection.

SIEM takes this even further. It acts as a centralized brain that collects logs and alerts from many sources: NGFWs, XDRs, HIDS (Host-based Intrusion Detection Systems), NIDS (Network-based IDS), HIPS, NIPS, antivirus, email gateways, and more. SIEM platforms organize this data, look for patterns, and can even trigger automated alerts when something suspicious happens.

Together, these tools create a layered defense. NGFWs and detection tools monitor and block threats, XDR pulls it all together across systems, and SIEM keeps historical records, correlation logic, and alerting rules to give security teams deep visibility.

Learning how these systems interact showed me that cybersecurity is like a team sport—no tool works alone. Real protection comes from making them work together and seeing the bigger picture.

## Project 8: Wireshark, the Network Investigation Superhero!

Wireshark is like a magnifying glass for your network—it lets you see exactly what's happening in the background as data travels between devices. It's one of the most powerful tools in cybersecurity and IT because it captures and analyzes real-time network traffic, down to the smallest detail.

Whether you're troubleshooting a slow connection, investigating a security alert, or just learning how packets work, Wireshark shows it all: source and destination IPs, ports, protocols, payload data, even handshake steps in protocols like TCP or TLS.

You can use filters in Wireshark to narrow down exactly what you're looking for—maybe just HTTP traffic, or maybe just packets to or from a suspicious IP. You can even follow entire TCP streams to reconstruct conversations between a client and server.

In security investigations, Wireshark helps spot things like:

- Failed login attempts
- Unusual DNS requests
- Malware trying to call home
- Data being sent in plain text

It's also great for verifying encryption. For example, if you're testing HTTPS, you can use Wireshark to confirm that the actual content is encrypted (you'll just see gibberish in the payload section).

Learning to use Wireshark gave me confidence to dig deeper into the traffic on any network and trace the exact path of a problem. It's like having X-ray vision for packets—every cybersecurity analyst should have it in their toolbox.

## Project 9: Digital Forensics: Finding Clues in the Cyber Crime Scene

Digital forensics is the process of collecting, analyzing, and preserving digital evidence after a security incident. It's like being a cyber detective—looking for clues in hard drives, logs, memory, and network traffic to figure out what happened, how it happened, and who was behind it.

In cybersecurity, digital forensics is used after things go wrong. Whether it's a malware outbreak, insider threat, or data breach, forensics helps piece together the digital footprint the attacker left behind.

The process usually includes several steps:

- Identification – spotting suspicious activity or compromised systems.
- Preservation – making a copy (image) of the data to avoid tampering.
- Analysis – digging into files, system logs, registry data, and timelines.
- Reporting – documenting findings for legal, internal, or security use.

Forensic tools like FTK, Autopsy, and EnCase allow investigators to recover deleted files, examine browser history, extract USB device connections, and more. Even something as small as a timestamp on a file can be a critical clue.

Learning about digital forensics helped me appreciate how detailed and careful cybersecurity investigations need to be. It's not just about stopping threats—it's also about understanding them well enough to prevent the next one and hold attackers accountable.

## Project 10: Incident Response: What Happens After the Alarm Goes Off

Incident response is the process an organization follows after discovering a cybersecurity incident—like a malware infection, data breach, or phishing attack. It's all about acting fast to minimize damage, stop the spread, and get operations back to normal.

The standard incident response lifecycle includes six phases:

1. Preparation – Set up policies, tools, and team roles before anything happens.
2. Identification – Detect the incident and determine its nature and scope.
3. Containment – Isolate affected systems to prevent further harm.
4. Eradication – Remove the threat (e.g., delete malware, patch vulnerabilities).
5. Recovery – Restore systems to a clean state and monitor for re-infection.
6. Lessons Learned – Review what happened and improve future response plans.

Effective incident response depends on having the right tools and communication. Analysts often use SIEMs to detect unusual activity, EDR/XDR platforms to investigate endpoints, and forensics tools to understand what the attacker did.

During an active incident, teams must act quickly but carefully—especially when attackers may still be active on the network. It's also critical to notify leadership, legal teams, and possibly customers if data is involved.

Learning about incident response taught me that cybersecurity isn't just about prevention—it's about how you react when things go wrong. A fast, coordinated response can be the difference between a small issue and a full-blown disaster.

## Project 11: CompTIA A+ IT Fundamentals

The CompTIA A+ certification is built around core principles that define the role of an entry-level IT professional. These principles span multiple domains of technical knowledge and practical support skills, ensuring a well-rounded foundation in computing environments. A fundamental principle of A+ is the understanding of computer hardware, which includes components such as motherboards, processors, RAM, power supplies, and storage drives. Professionals must be able to identify, install, configure, and troubleshoot these parts to ensure optimal performance and compatibility.

Another key area is operating systems, primarily Windows, but also including Linux, macOS, and mobile platforms like iOS and Android. The A+ framework covers installation procedures, file system structures, command-line tools, system utilities, and user management. Understanding these systems allows technicians to support a wide variety of environments and perform system recovery, upgrades, and maintenance.

Networking is also a central focus. CompTIA A+ ensures knowledge of IP addressing, DNS, DHCP, network types (LAN, WAN, PAN), and wireless standards such as 802.11 protocols. It emphasizes the ability to identify and troubleshoot connectivity issues using tools like ping, ipconfig, tracert, and nslookup. Basic knowledge of routers, switches, and network cabling is essential to support both home and small business networks.

Troubleshooting, both for software and hardware, is at the heart of the A+ approach. It emphasizes a structured methodology: identify the problem, establish a theory, test the theory, implement a solution, verify full system functionality, and document the findings. This principle supports effective problem-solving across scenarios such as system crashes, driver conflicts, overheating hardware, or unresponsive devices.

Security is a foundational principle that weaves through every domain. CompTIA A+ emphasizes both digital and physical security, including the use of antivirus tools, firewalls, encryption, access controls, and user education to combat threats like phishing, malware, and social engineering. Technicians are expected to apply secure practices when handling sensitive data or managing user access.

The certification also includes mobile devices and their unique requirements. Knowledge of mobile operating systems, synchronization processes, connection issues, and device management are all part of the skill set. Virtualization and cloud computing are likewise essential in modern IT, with emphasis on understanding cloud models (IaaS, SaaS, PaaS), virtual machines, and syncing data between devices using services like OneDrive or Google Drive.

Finally, operational procedures tie all technical skills together with safety, professionalism, and compliance. This includes proper documentation, adherence to service-level agreements, safe disposal of hardware and data, and best practices in communication and customer service. CompTIA A+ ensures technicians follow industry protocols for safety, handle issues with professionalism, and maintain high standards in workplace conduct.

In summary, the CompTIA A+ principles ensure that IT professionals are equipped with broad, essential knowledge across hardware, software, networking, troubleshooting, security, mobile technology, cloud services, and operational conduct. This foundation prepares individuals for real-world support roles in modern computing environments.

## Project 12: ITIL -IT And Business Integration

The ITIL 4 framework represents the most up-to-date version of the globally recognized standard for IT service management (ITSM). It is designed to align IT services with the needs of the business, focusing on value co-creation, digital transformation, and service-centric thinking. At its core, ITIL 4 introduces a more flexible and holistic approach to managing services, emphasizing collaboration, automation, and continual improvement.

A central component of ITIL 4 is the Service Value System (SVS), which illustrates how various components and activities work together to facilitate value creation. The SVS includes five key elements: guiding principles, governance, the service value chain, continual improvement, and practices. Together, these components ensure that the organization remains adaptable and focused on delivering measurable outcomes for stakeholders.

The seven guiding principles—such as “focus on value,” “start where you are,” and “progress iteratively with feedback”—are fundamental behaviors that influence decision-making and actions across the IT organization. These principles are applicable at all levels and in all roles, promoting a culture of transparency, agility, and efficiency.

The service value chain is another essential concept, offering a flexible operating model that outlines six key activities: plan, improve, engage, design & transition, obtain/build, and deliver & support. These activities can be combined in various ways to form value streams, depending on the organization’s service delivery and support needs. This approach allows IT teams to adapt their workflows to fit both traditional and agile development environments.

ITIL 4 replaces the previous concept of “processes” with 34 management practices that cover a wide range of capabilities, from incident management and change enablement to service level management and risk management. These practices ensure that organizations have the necessary tools and methods to deliver high-quality services consistently. Each practice is designed to be adaptable and scalable, allowing integration with modern tools, DevOps pipelines, and cloud-based platforms.

Continual improvement is not a one-time task but a mindset woven throughout ITIL 4. The continual improvement model provides a structured approach for evaluating current states, defining measurable goals, and planning changes that drive service performance and business alignment over time. This ensures that IT remains responsive to internal and external changes, regulatory shifts, and evolving customer expectations.

Furthermore, ITIL 4 recognizes the importance of governance and risk in the broader business landscape. By defining decision-making structures and oversight mechanisms, organizations can ensure accountability and compliance while supporting innovation and operational excellence.

In essence, ITIL 4 is more than just a set of best practices; it is a dynamic framework that bridges the gap between IT operations and business strategy. It empowers organizations to manage complex IT environments while maintaining a relentless focus on delivering value, reducing waste, and continually evolving to meet modern demands.

## Project 13: Cloud and the Remote Virtual Network

Modern IT infrastructure is increasingly built on virtualization and remote access technologies, enabling greater scalability, flexibility, and security. At the heart of this evolution is cloud computing, which delivers computing resources over the internet on-demand. Cloud environments can be public, private, or hybrid, and they are typically offered through service models such as Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). These models allow organizations to shift from traditional on-premise data centers to scalable, cost-effective virtual infrastructure that adjusts to business needs in real time.

A fundamental component enabling cloud services is the hypervisor. Hypervisors are virtualization platforms that allow multiple virtual machines (VMs) to run on a single physical host. There are two types: Type 1 hypervisors (bare-metal), such as VMware ESXi and Microsoft Hyper-V, run directly on the hardware; Type 2 hypervisors, such as Oracle VirtualBox and VMware Workstation, run on a host operating system. These tools abstract hardware resources, enabling the creation of isolated virtual servers that can host different operating systems and applications on the same physical device.

Virtual servers, often provisioned within a cloud environment or through on-prem virtualization, provide the same capabilities as traditional physical servers but with more flexibility. They can be quickly deployed, duplicated, migrated, and scaled as needed. Virtual servers reduce hardware costs and increase efficiency while supporting high availability and disaster recovery strategies. Each VM runs its own OS and can be used for roles such as web hosting, application delivery, or database management.

Security and accessibility are critical in virtualized environments. Virtual Private Networks (VPNs) play a crucial role by creating encrypted tunnels between remote devices and internal networks. VPNs ensure that users working offsite can securely access organizational resources over the internet as if they were physically connected to the internal network. This is especially important in hybrid work environments, where secure access to sensitive data must be maintained without exposing it to the open internet.

Remote Desktop Protocol (RDP) further extends remote access capabilities by allowing users to connect to and control a desktop or server from another device. Commonly used in IT support and remote work scenarios, RDP enables full GUI access to a remote system, including applications, files, and networked drives. While convenient, RDP must be secured using strong authentication, network-level protection, and ideally through VPNs or gateway services to prevent unauthorized access.

Together, these technologies form the foundation of modern IT ecosystems. Cloud computing provides the backbone for flexible infrastructure. Hypervisors enable efficient use of hardware through virtualization. Virtual servers replace static physical machines with dynamic, scalable systems. VPNs and RDP ensure secure, seamless remote access. Mastery of these components is essential for supporting today's distributed, digital-first environments.

## Project 14: Ports, Ports and MORE PORTS!!!!

In the world of networking and cybersecurity, understanding common ports and their associated protocols is essential for diagnosing issues, securing networks, and configuring firewalls. Each port is linked to a specific protocol and service, allowing devices to communicate over the internet or internal networks with precision and consistency.

Port 80 is one of the most widely recognized ports, associated with HTTP (Hypertext Transfer Protocol). It is used for unencrypted web traffic, allowing browsers to request and receive web pages from servers. In contrast, port 443 is tied to HTTPS (Hypertext Transfer Protocol Secure), which encrypts traffic using SSL/TLS, protecting data in transit and ensuring secure communication between users and websites.

For email services, several ports are commonly used depending on the protocol and function. Port 25 is associated with SMTP (Simple Mail Transfer Protocol) for sending emails between mail servers. Port 110 is used for POP3 (Post Office Protocol v3), which downloads emails from a server to a local device and typically deletes them from the server. Port 143 supports IMAP (Internet Message Access Protocol), allowing users to view and manage email messages directly on the mail server without downloading them.

File transfer operations rely on ports like 21, used by FTP (File Transfer Protocol) for transferring files between systems, often with minimal security. Port 22 is used by SSH (Secure Shell), which provides secure command-line access to remote systems. It also underpins secure file transfers through SFTP, which operates over SSH. Port 23 is assigned to Telnet, a legacy protocol similar to SSH but without encryption, making it insecure for modern use.

Port 53 is associated with DNS (Domain Name System) and serves both TCP and UDP protocols. It resolves domain names into IP addresses, allowing users to connect to websites using human-readable names instead of numerical addresses. Port 67 and 68 are used by DHCP (Dynamic Host Configuration Protocol) to automatically assign IP addresses to devices on a network.

Remote desktop services utilize port 3389, which corresponds to RDP (Remote Desktop Protocol). This allows remote graphical access to a Windows system and is widely used for IT support and administration. Port 389 is tied to LDAP (Lightweight Directory Access Protocol), used for querying and modifying directory services like Microsoft Active Directory.

Secure communications across the web and email often require encryption. Port 993 supports IMAPS, a secure version of IMAP, while port 995 is used for POP3S, the encrypted form of POP3. Port 465 is sometimes used for SMTPS, an encrypted version of SMTP, though modern systems typically use port 587 for sending encrypted emails.

Other notable ports include port 69 for TFTP (Trivial File Transfer Protocol), a simplified, unsecured file transfer tool often used for firmware updates or booting devices; port 161 for SNMP (Simple Network Management Protocol), used for monitoring and managing networked devices; and port 137–139 and 445 for NetBIOS/SMB, used in Windows file sharing and printer access.

Mastering these common ports and their purposes is essential for configuring firewalls, conducting vulnerability scans, interpreting network traffic, and ensuring secure and efficient communication between systems.

## Project 15: Locked Down

Encryption is a fundamental pillar of modern cybersecurity, ensuring that data remains confidential and secure during storage and transmission. At its core, encryption is the process of converting readable information, known as plaintext, into an unreadable format, called ciphertext. This transformation is done using mathematical algorithms and cryptographic keys. Only someone with the correct decryption key can convert the ciphertext back into its original plaintext form, protecting the data from unauthorized access.

There are two main types of encryption: symmetric and asymmetric. Symmetric encryption uses a single shared key for both encryption and decryption. This method is fast and efficient, making it ideal for encrypting large amounts of data such as files, databases, or VPN tunnels. However, the challenge lies in securely distributing the key between parties. Common symmetric encryption algorithms include AES (Advanced Encryption Standard) and DES (Data Encryption Standard).

Asymmetric encryption, on the other hand, uses a pair of keys: a public key and a private key. The public key is freely distributed and used to encrypt data, while the private key is kept secret and used to decrypt it. This model solves the problem of key distribution and forms the backbone of secure web communications, such as HTTPS. RSA is one of the most widely used asymmetric encryption algorithms, often used to securely exchange symmetric keys before switching to faster symmetric encryption for the rest of the communication.

In real-world applications, encryption protects sensitive data in many environments. For example, when accessing a secure website, the browser and web server perform a handshake to exchange keys and establish a secure HTTPS connection. This ensures that any information sent between them—such as passwords, payment details, or personal data—is encrypted and safe from eavesdroppers. Similarly, full-disk encryption tools like BitLocker or FileVault protect data stored on hard drives, preventing access if the device is lost or stolen.

Encryption also supports digital signatures, ensuring data integrity and authenticity. When a message is signed with a private key, anyone with the sender's public key can verify that the message was not altered and originated from the expected source. This is crucial in fields like software distribution, where verifying the legitimacy of updates and installers is essential.

Key management is another critical aspect of encryption. Without proper handling, secure algorithms can be undermined by weak key generation, poor storage practices, or unauthorized key access. Organizations typically use certificate authorities (CAs) and public key infrastructure (PKI) to manage and validate keys across networks and users.

In summary, encryption is not just a single tool but a comprehensive strategy involving algorithms, key management, and best practices. It protects confidentiality, ensures data integrity, and builds trust across digital systems, forming the backbone of secure communication in the modern world.

## Project 16: SSH and Telnet

SSH (Secure Shell) and Telnet are both tools used to remotely access and control computers, especially servers. Think of them like remote keyboards — they let someone type commands into a machine that could be across the room or across the world. The difference lies in how securely they do it.

Telnet was one of the first remote command tools used in networking. It connects to a device (like a router, switch, or server) using port 23, and once connected, allows users to type and execute commands as if they were sitting in front of it. However, Telnet sends all data, including usernames and passwords, in plain text. That means if someone is watching the network (with a tool like Wireshark), they could see everything being typed. Because of this, Telnet is considered outdated and insecure for most modern environments.

To set up Telnet, the service must be enabled on the target device, and a Telnet client (like the Windows Command Prompt with `telnet IPaddress`) is used to connect. It's often disabled by default today for security reasons.

SSH, on the other hand, does the same job as Telnet but in a much safer way. It connects over port 22 and encrypts all traffic between the two devices. That means even if someone intercepts the data, it would appear scrambled and unreadable. SSH also supports stronger authentication methods, like using cryptographic key pairs instead of just passwords.

To set up SSH, an SSH server like OpenSSH needs to be running on the device. On most Linux systems, SSH is installed and running by default. From a client machine, a user can simply open a terminal and type a command like `ssh username@IPaddress`. On Windows, PowerShell or applications like PuTTY are commonly used to initiate SSH sessions.

Today, SSH is the industry standard for remote access because it protects login credentials and command outputs from eavesdropping. It's commonly used by system administrators, developers, and network engineers to manage systems securely.

In summary, Telnet is fast but unsafe, while SSH is secure and reliable. For any serious IT work, SSH is the go-to tool for remote access and management.

Ethical hacking, also called penetration testing or white-hat hacking, is the practice of legally and intentionally probing systems, networks, or applications for vulnerabilities—before malicious hackers can find them. Ethical hackers use the same tools and techniques as real attackers, but their goal is to fix the problems, not exploit them.

One of the core goals of ethical hacking is to simulate real-world attacks. This means understanding how cybercriminals think and operate. Attackers can use many entry points, called attack vectors, to break into systems. These include weak passwords, misconfigured firewalls, outdated software, social engineering, open ports, and even physical access.

Some of the most common types of attacks include:

- Phishing – Fake emails or websites that trick users into revealing credentials.
- Malware – Malicious software like ransomware, trojans, and worms.
- Brute-force attacks – Guessing passwords by trying many combinations.
- SQL Injection – Inserting code into a web form to access backend databases.
- Denial of Service (DoS) – Flooding a system to make it unavailable.

Ethical hackers often encounter various malware tools used by threat actors such as keyloggers, remote access trojans (RATs), botnets, and spyware. Part of penetration testing is understanding how these work, how to recognize their behavior, and how to remove or block them.

Insider threats are also a major concern. These are attacks that come from within the organization—employees or contractors who misuse access intentionally or unintentionally. Because insiders already have access, they can be harder to detect and more damaging if not monitored properly.

Phishing is one of the most successful attack vectors because it relies on human error. Signs of phishing include poor grammar, suspicious links, emails from unknown senders, and urgent language that pressures the user. Cybersecurity professionals are trained to recognize these signs and educate employees through simulated phishing campaigns.

Before conducting a full penetration test, ethical hackers usually start with a vulnerability scan to identify potential weaknesses. These scans can be done with tools like Nessus, OpenVAS, or Nmap. Here's an example command:

```
nmap -sV -T4 192.168.1.0/24
```

This command scans a subnet for open ports and tries to determine what services and versions are running.

Another tool often used is Nikto for scanning web servers:

```
nikto -h http://targetwebsite.com
```

For password testing, tools like Hydra or John the Ripper are used:

```
hydra -l admin -P /usr/share/wordlists/rockyou.txt 192.168.1.100 ssh
```

These tools help simulate real attacks and test how strong the target's defenses are.

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### Step-by-Step Example: Basic Penetration Test Workflow

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1. Reconnaissance – Gather info about the target (domains, IPs, OS).

```
whois targetdomain.com
```

```
nslookup targetdomain.com
```

2. Scanning – Find open ports and services.

```
nmap -sS -Pn targetip
```

3. Enumeration – Pull detailed data from services.

```
enum4linux targetip
```

4. Exploitation – Try to exploit vulnerabilities.

```
searchsploit vsftpd
```

```
msfconsole
```

5. Privilege Escalation – Gain admin access.

linux-exploit-suggester.sh

6. Post-Exploitation – Access sensitive data, create persistence.

netstat -tulnp

crontab -e

7. Reporting – Document findings, include recommendations.

Learning ethical hacking gave me the mindset of an attacker, which made me a stronger defender. It's not about being destructive—it's about anticipating what bad actors might do and building systems that are resilient, monitored, and secured against those threats.